

Yotam Sechayk

Ph.D. Candidate · The University of Tokyo · Human-Computer Interaction & Accessibility
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Research Interests

I am a human-computer interaction and accessibility researcher with an interdisciplinary background. My work focuses on **making visual communication visually accessible** for low-vision people. I study how low-vision people use their residual vision to access lecture videos, data charts, and the web. I follow user-centered and ability-based design, working with low-vision people throughout the design process and building on the abilities they have. As a low-vision person myself, I design interaction techniques and AI-powered systems that *enhance*, rather than replace, existing visual strategies. I have published **six full papers** at CHI, ASSETS, and UIST, and my work has received **three paper awards**, including an ASSETS Honorable Mention.

Education

- 2024.04 – Present **The University of Tokyo** · Tokyo, Japan
Ph.D. in Creative Informatics, Graduate School of Information Science and Technology (expected 2027.03)
Advisor: Prof. Takeo Igarashi · Co-advisor: Prof. Ariel Shamir (Reichman University)
Dissertation (in progress): *Visual Communication for Low-Vision Information Access*
- 2022.04 – 2024.03 **The University of Tokyo** · Tokyo, Japan
M.Sc. in Creative Informatics, Graduate School of Information Science and Technology
Advisor: Prof. Takeo Igarashi
Thesis: *Smart Replay: Visual-Temporal Accessibility in e-learning Videos*
- 2017.04 – 2017.09 **Waseda University** · Tokyo, Japan
Exchange Student, School of International Liberal Studies
- 2014.10 – 2018.09 **Tel Aviv University** · Tel Aviv, Israel
B.Sc., Double Major in Computer Science and Film & Television
Graduated with honors (magna cum laude; GPA 93/100).

Research Experience

- 2022.04 – Present **Igarashi Lab** · The University of Tokyo
Advisor: Prof. Takeo Igarashi · Co-advisor: Prof. Ariel Shamir (Reichman University)
- Lead a SoftBank Beyond AI CIP project on web accessibility with AI agents for low-vision users (2026 – present). Manage the research budget, supervise a research intern, and coordinate development with the SoftBank team.
 - Built *Improving Low-Vision Chart Accessibility via On-Cursor Visual Context*, comparing two pointer-based context methods (Dynamic Context, Mini-map) with $N=22$ LVI. [C.5]
 - Designed *VeasyGuide*, a tool that detects instructor actions in lecture videos and dynamically highlights and magnifies them for low-vision learners. Evaluated with 8 LV and 8 sighted participants. Open source and available online at veasyguide.github.io/app. [C.3]

- Earlier and smaller projects on visual-temporal accessibility of e-learning videos, chart reading, and creative tools: *Graph Guide* [D.4], *ShowMe* [D.3], *SmartLearn* [W.1], *Smart Replay* [D.1], and *MyStoryKnight* [D.2].

2024.08 – 2024.10 **University of Texas at Austin** · Austin, TX
 Visiting Researcher
 Advisor: Prof. Amy Pavel
 Recruited participants and ran the user studies for *VeasyGuide*. [C.3] Continued the collaboration after the visit on projects including *Task Mode*. [C.2]

2023.10 – 2024.03 **Honda Research Institute Japan** · Saitama, Japan
 Research Intern, Human-Robot Interaction
 Supervisor: Dr. Christian Arzate Cruz
 Augmented facial-expression data with 3D morphable models to improve arousal–valence prediction in HRI settings. [C.1]

Work Experience

2018.03 – 2020.10 **Cadence Design Systems** · Israel
 Software Engineer, Multicore Engine Team
 Developed semiconductor verification tools utilizing parallel computing.

2015.03 – 2017.03 **Tel Aviv University** · Tel Aviv, Israel
 Head of Technology, Disability Support
 Investigated software and hardware solutions to assist students with disabilities and supported their adoption in coursework.

Publications

Seven full papers at international venues (CHI, ASSETS, UIST, RO-MAN), a UIST Doctoral Symposium paper, eight posters, demos, and workshop papers, and four papers at Japanese venues (WISS, INTERACTION). Three papers received awards, including an ASSETS Honorable Mention and a WISS Best Paper Award.

* indicates equal contribution. My name is underlined. Entries are numbered chronologically (1 = earliest).

International Conference Papers

C.7 **LandmarkLens: Predicting and Presenting Effective Landmarks for Mixed-Reality Urban Exploration**

Chu Li, Yotam Sechayk, Jared Hwang, Jon E. Froehlich, Takeo Igarashi
 UIST 2026: ACM Symposium on User Interface Software and Technology.

C.6 **Ups and Downs: Low-Vision Perspective on Smart Elevators and Industry Design Practices**

Masaki Sawada, Yotam Sechayk, Manabi Miyagi
 ASSETS 2026: ACM SIGACCESS Conference on Computers and Accessibility (experience report).

C.5 **Improving Low-Vision Chart Accessibility via On-Cursor Visual Context**

Yotam Sechayk, Hennes Rave, Max Rädler, Mark Colley, Zhongyi Zhou, Ariel Shamir, Takeo Igarashi
 CHI 2026: ACM Conference on Human Factors in Computing Systems.

- C.4 No Pixel Left Behind: Filling Gaps in Anime Colorization**
Masahiro Kono, Akinobu Maejima, Yuki Koyama, Yotam Sechayk, Takeo Igarashi
CHI 2026: ACM Conference on Human Factors in Computing Systems.
- C.3 VeasyGuide: Personalized Visual Guidance for Low-vision Learners on Instructor Actions in Presentation Videos**
Yotam Sechayk, Ariel Shamir, Amy Pavel, Takeo Igarashi
ASSETS 2025: ACM SIGACCESS Conference on Computers and Accessibility.
 **Honorable Mention Award**
- C.2 Task Mode: Dynamic Filtering for Task-Specific Web Navigation using LLMs**
Ananya Gubbi Mohanbabu, Yotam Sechayk, Amy Pavel
ASSETS 2025: ACM SIGACCESS Conference on Computers and Accessibility.
- C.1 Data Augmentation for 3DMM-based Arousal-Valence Prediction for HRI**
Christian Arzate Cruz*, Yotam Sechayk*, Takeo Igarashi, Randy Gomez
RO-MAN 2024: IEEE International Conference on Robot and Human Interactive Communication.

Doctoral Symposium

- DS.1 Visual Communication for Low-Vision Information Access: From Personalized Highlights to AI Agents**
Yotam Sechayk
UIST Adjunct 2026: Adjunct Proceedings of the ACM Symposium on User Interface Software and Technology.

Workshops, Posters, Demos, and Extended Abstracts

- W.8 Cyber-Physical Systems for Accessibility and Ability Augmentation: Bridging Diverse Communities**
Shuchang Xu*, Riku Arakawa*, Mina Huh*, Nandi Zhang*, Tianyu Zhang*, Wazeer Zulfikar*, Ruei-Che Chang*, Yotam Sechayk*, Huamin Qu, Amy Pavel, Franklin Mingzhe Li, Yukang Yan, Brian A. Smith, Pattie Maes
UIST Adjunct 2026: Adjunct Proceedings of the ACM Symposium on User Interface Software and Technology (workshop proposal).
- W.7 PreView: Rehearsing Real Routes in VR to Support Independent Navigation with Low-Vision**
Yotam Sechayk, Xinyun Cao, Dhruv Jain, Takeo Igarashi
UIST Adjunct 2026: Adjunct Proceedings of the ACM Symposium on User Interface Software and Technology (demo).
- W.6 See What I Mean: A Technology Probe of AI Assistants for Chart Exploration with Low Vision**
Yotam Sechayk, Lucy Jiang, Yuhang Zhao, Takeo Igarashi
ASSETS 2026: ACM SIGACCESS Conference on Computers and Accessibility (poster).
- W.5 Exploring the Role of User Comments Throughout the Stages of Video-Based Task-Learning**
Nayoung Kim*, Yotam Sechayk*, Zhongyi Zhou, Takeo Igarashi
CHI EA 2026: Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems (poster).

- W.4 A Longitudinal Autoethnography of Email Access for a Professional with Chronic Illness and ADHD: Preliminary Insights**
Veronica Pimenova, Yotam Sechayk, Fabricio Murai, Andrew Hundt, Shiri Dori-Hacohen
ASSETS 2025: ACM SIGACCESS Conference on Computers and Accessibility (poster).
- W.3 Confidence Estimation of Few-shot Patch-based Learning for Anime-style Colorization**
Yuexiang Ji, Akinobu Maejima, Yotam Sechayk, Yuki Koyama, Takeo Igarashi
SIGGRAPH 2025 Posters.
- W.2 ImprovMate: Multimodal AI Assistant for Improv Actor Training**
Riccardo Drago, Yotam Sechayk, Mustafa Doga Dogan, Andrea Sanna, Takeo Igarashi
DIS 2025 Companion: ACM Conference on Designing Interactive Systems (work-in-progress).
- W.1 SmartLearn: Visual-Temporal Accessibility for Slide-based e-learning Videos**
Yotam Sechayk, Ariel Shamir, Takeo Igarashi
CHI EA 2024: Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems (late-breaking work).

Domestic Conference Papers (Japan)

- D.4 Graph Guide: 低視力者支援のためのセマンティック Focus+Context グラフ表示**
Graph Guide: Semantic Focus+Context Graph Display for Low-Vision Users
Yotam Sechayk, Yuan Li, Hennes Rave, Mark Colley, Ariel Shamir, Takeo Igarashi
WISS 2025: 第 33 回インタラクティブシステムとソフトウェアに関するワークショップ, IPSJ (in Japanese).
 **Best Paper Award (最優秀論文賞)**
- D.3 ShowMe: 対話的な強調表示と拡大表示によるプレゼンテーションビデオの視覚的アクセシビリティの改善**
ShowMe: Interactive Highlighting and Magnification for Improving Visual Accessibility of Presentation Videos
Yotam Sechayk, Ariel Shamir, Takeo Igarashi
WISS 2024: 第 32 回インタラクティブシステムとソフトウェアに関するワークショップ, IPSJ (in Japanese).
- D.2 MyStoryKnight: A Character-drawing Driven Storytelling System Using LLM Hallucinations**
Yotam Sechayk, Gabriela A. Penarska, Ingrid A. Randsalu, Christian Arzate Cruz, Takeo Igarashi
INTERACTION 2024, IPSJ (poster/demo).
- D.1 Smart Replay: e ラーニング動画における視覚的・時間的アクセシビリティの向上**
Smart Replay: Improving Visual and Temporal Accessibility in E-learning Videos
Yotam Sechayk, Ariel Shamir, Takeo Igarashi
WISS 2023: 第 31 回インタラクティブシステムとソフトウェアに関するワークショップ, IPSJ (in Japanese).
 **Helpfeel Award for Corporate Excellence**

Patents

- Published **Data Collection Method and Data Collection Device**
Christian Arzate Cruz, Randy Gomez, Yotam Sechayk
WO 2026/053487 A1 (PCT/JP2025/015777), published March 2026. Assignee: Honda Motor Co., Ltd.
Invented during internship at Honda Research Institute Japan. **[C.1]**

Awards & Honors

- 2025 **Best Paper Award** (*Saiyushu Ronbunsho*) · WISS 2025
Awarded for Graph Guide. [D.4]
- 2025 **Honorable Mention Award** · ACM ASSETS 2025
Awarded for VeasyGuide. [C.3]
- 2023 **Helpfeel Award for Corporate Excellence** · WISS 2023
Awarded for Smart Replay. [D.1]
- 2018 **Graduated with Honors** (*magna cum laude*) · Tel Aviv University

Grants & Fellowships

Co-wrote three funded research proposals that secured about USD 225,000 (JPY 35,500,000) for the lab. Received two MEXT scholarships worth about USD 80,000 (JPY 12,800,000) in tuition and stipends.

- 2024 – Present **MEXT Scholarship for Ph.D. Studies in Japan** · Japanese Ministry of Education, Culture, Sports, Science and Technology (MEXT)
Full tuition and a monthly stipend of JPY 148,000 for the doctoral program (about USD 49,000 in total).
- 2026 – 2027 **Softbank Beyond AI CIP Grant** · Softbank
JPY 20,000,000 (about USD 127,000) for web accessibility with AI agents (Jun 2026 – Mar 2027).
Co-wrote the proposal; project lead. PI: Prof. Takeo Igarashi.
- 2025 **Softbank Beyond AI Research Grant** · Softbank
JPY 10,000,000 (about USD 63,000). Co-wrote the proposal. PI: Prof. Takeo Igarashi.
- 2023 **Google Award for Inclusion Research** · Google
USD 35,000 for “Enhance E-Learning Accessibility: Exploring AI Supported Temporal Approaches for Inclusive Online Education.” Co-wrote the proposal; PI: Prof. Takeo Igarashi.
- 2022 – 2024 **MEXT Scholarship for Master’s Studies in Japan** · Japanese Ministry of Education, Culture, Sports, Science and Technology (MEXT)
Full tuition and a monthly stipend of JPY 146,000 for the master’s program (about USD 32,000 in total).

Invited Talks & Research Visits

Invited talks at KAIST, Keio University, Hokkaido University, Miraikan, UT Austin, and Microsoft Israel, and research visits to University College London, the University of Washington, and UT Austin.

- 2026.04 **Research Visit** · University College London (UCL), London, UK
Host: Dr. Mark Colley (UCL Interaction Centre).
- 2026.02 **Increasing Access in Online Education through Personalized Visual Highlights**
Invited Talk, KIXLAB, KAIST, Daejeon, South Korea. Host: Prof. Juho Kim.
- 2026.01 **Increasing Access in Online Education through Personalized Visual Highlights**
Invited Talk, Embodied Media Project, Keio University Graduate School of Media Design (KMD), Yokohama, Japan. Host: Dr. Takatoshi Yoshida.
- 2025.12 **Increasing Access in Online Education through Personalized Visual Highlights**
Invited Talk, HCI Lab, Hokkaido University, Sapporo, Japan. Host: Prof. Daisuke Sakamoto.
- 2025.11 **Highlights and Context: Augmenting Sight for Low-vision People**
Invited Talk, Accessibility Lab, Miraikan (National Museum of Emerging Science and Innovation), Tokyo, Japan. Host: Dr. Hironobu Takagi.

- 2025.09 – 2025.11 **Research Visit** · University of Washington, Seattle, WA
Host: Prof. Jon E. Froehlich (Makeability Lab).
- 2024.10 **Awareness-(less) based Accessibility**
Invited Talk, University of Texas at Austin. Host: Prof. Amy Pavel.
- 2024.08 – 2024.10 **Research Visit** · University of Texas at Austin, Austin, TX
Host: Prof. Amy Pavel.
- 2023.05 **Accessibility and Usability in Online Learning Videos**
Invited Talk, Microsoft Israel. Host: Deep Learning Team.

Conference Presentations

- 2026 ACM CHI 2026 [C.5] · ACM ASSETS 2026 [C.6] (joint) · ACM UIST 2026 Doctoral Symposium [DS.1]
- 2025 ACM ASSETS 2025 [C.3] · WISS 2025 [D.4]
- 2024 WISS 2024 [D.3]
- 2023 WISS 2023 [D.1]

Teaching

- 2026 – Present **Creative Informatics Seminar (Master's and Doctoral)**
Teaching Assistant · Graduate School of Information Science and Technology, The University of Tokyo (4860-2008/2009)
Led a complete restructuring of the course format, backed by student feedback and prior research on interdisciplinary research and research communication. The new format launched in April 2026.
- Introduced weekly poster sessions to give students more opportunities to interact.
 - Created online mechanisms for students to give each other feedback.
 - Received positive faculty and student feedback and strong participation.
- 2026 **Accessibility in HCI and the AI Revolution**
Guest Lecture · Undergraduate Human-Computer Interaction (0510071), The University of Tokyo
Instructor of record: Prof. Takeo Igarashi.
- 2025 **Accessibility in HCI and the AI Revolution**
Guest Lecture · Undergraduate Human-Computer Interaction (0510071), The University of Tokyo
Instructor of record: Prof. Takeo Igarashi.

Mentoring & Advising

Mentored ten students and junior researchers, from undergraduates to Ph.D. students, at institutions in Japan, the US, Israel, Italy, and Korea. Five mentees co-authored papers from this work.

- 2026 – present **Faith Luo** · Research Intern on web accessibility with AI agents (Software Engineer, thatgamecompany)
- 2026 – present **Ritsu Ueda** · Georgia Institute of Technology
- 2026 – present **Masaki Sawada** · Tsukuba University of Technology. [C.6]
- 2026 **Koyo Tanaka** · University of California, San Diego

- 2025 – present **Eleyah Melamed** · Reichman University
- 2025 – present **Xinyun Cao** · University of Michigan. [W.7]
- 2024 – present **Veronica Lewis** · Ph.D., George Mason University
- 2024 – present **Veronica Pimenova** · University of Michigan. [W.4]
- 2024 – 2025 **Riccardo Drago** · M.Sc., Polytechnic University of Turin. [W.2]
- 2024 **Nayoung Kim** · Undergraduate, DGIST. [W.5]

Academic Service

Organizing

- 2026 – 2027 **Accessibility Chair** · ACM CHI 2027 Organizing Committee

Reviewing

- 2026 ACM CHI 2026 · ACM UIST 2026 · ACM MobileHCI 2026
 - 2025 ACM CHI 2025 · ACM UIST 2025 · ACM DIS 2025 · ACM IDC 2025
 - 2024 ACM CHI 2024 · ACM UIST 2024
- Special Recognition for Outstanding Reviews: CHI 2025 (×2), UIST 2025, CHI 2026.

Outreach & Media

- 2026.07 **School Visit** · Tsukuba Special Needs Education School for the Visually Impaired
Introduced our educational accessibility tools to teachers and discussed collaboration, with Dr. Mika Oki and Prof. Takeo Igarashi. July 28, 2026.
- 2026.05 **The University of Tokyo Visit for Low-Vision Students from Tsukuba University of Technology**
Organized and hosted four students with low vision and Prof. Makoto Kobayashi, including an introduction to our research. May 7, 2026.
- 2025 *AT and Me: An Interview with Yotam Sechayk*
Perkins School for the Blind · Paths to Technology · interview by Veronica Lewis, August 2025.

Languages

Hebrew (native) · English (fluent) · Japanese (advanced).